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Acacia

Acacia, commonly known as **wattles** or **acacias**, is a genus of about 1,084 species of shrubs and trees in the subfamily Mimosoideae of the pea family Fabaceae. Initially, it comprised a group of plant species native to Africa, South America, and Australasia, but is now reserved for species mainly from Australia, with others from New Guinea, Southeast Asia, and the Indian Ocean.[1] The genus name is Neo-Latin, borrowed from the Greek ἀκακία (*akakia*), a term used in antiquity to describe a preparation extracted from *Vachellia nilotica*, the original type species.

A number of species of *Acacia* have been introduced to various parts of the world, and two million hectares of commercial plantations have been established.[3]

Acacia	
Acacia baileyana Canberra	
Scientific classification	
Kingdom:	Plantae
Clade:	Tracheophytes
Clade:	Angiosperms
Clade:	Eudicots
Clade:	Rosids
Order:	Fabales
Family:	Fabaceae
Subfamily:	Caesalpinioideae
Clade:	Mimosoid clade
Genus:	*Acacia* Mill.[1]
Type species	
Acacia penninervis Sieber ex DC. (typ.cons.)[2]	
Species	

Description

Plants in the genus *Acacia* are shrubs or trees with bipinnate leaves, the mature leaves sometimes reduced to phyllodes or rarely absent. There are 2 small stipules at the base of the leaf, but sometimes fall off as the leaf matures. The flowers are borne in spikes or cylindrical heads, sometimes singly, in pairs or in racemes in the axils of leaves or phyllodes, sometimes in panicles on the ends of branches. Each spike or cylindrical head has many small golden-yellow to pale creamy-white flowers, each with 4 or 5 sepals and petals, more than 10 stamens, and a thread-like style that is longer than the stamens. The fruit is a variably-shaped pod, sometimes flat or cylindrical, containing seeds with a fleshy aril on the end.[4][5][6][7][8][9]

Taxonomy

The genus was first validly named in 1754 by Philip Miller in *The Gardeners Dictionary*.[10] In 1913 Nathaniel Lord Britton and Addison Brown selected *Mimosa scorpioides* L. (≡ *Acacia scorpioides* (L.) W.Wight = *Acacia nilotica* (L.) Delille), a species from Africa, as the lectotype of the name.

History

Genus *Acacia* was considered to contain some 1 352 species leading to 1986. That year, Leslie Pedley questioned the monophyletic nature of the genus, and proposed a split into three genera: *Acacia* sensu stricto (161 species), *Senegalia* (231 species) and *Racosperma* (960 species), the last name first proposed in 1829 by Carl Friedrich Philipp von Martius as the name of a section in *Acacia*,[11] but raised to generic rank in 1835.[12][13][14] In 2003, Pedley published a paper with 834 new combinations in *Racosperma* for species, most of which were formerly placed in *Acacia*.[15] All but 10 of these species are native to Australasia, where it constitutes the largest plant genus.[8]

In the early 2000s, it had become evident that the genus as it stood was not monophyletic and that several divergent lineages needed to be placed in separate genera. It turned out that one lineage comprising over 900 species mainly native to Australia, New Guinea, and Indonesia was not closely related to the much smaller group of African lineage that contained *A. nilotica* – the type species. This meant that the Australasian lineage (by far the most prolific in number of species) would need to be renamed. Pedley's proposed name of *Racosperma* for this group had received little acclaim in the botanical community. Australian botanists proposed a less disruptive solution, setting a different type species for *Acacia* (*A. penninervis*) and allowing this largest number of species to remain in *Acacia*, resulting in the two pan-tropical lineages being renamed *Vachellia* and *Senegalia*, and the two endemic American lineages renamed *Acaciella* and *Mariosousa*.[16]

In 2003, Anthony Orchard and Bruce Maslin filed a proposal to conserve the name *Acacia* with a different type, in order to retain the Australasian group of species in the genus *Acacia*.[14] Following a controversial decision to choose a new type for *Acacia* in 2005, the Australian component of *Acacia s.l.* now retains the name *Acacia*.[17][18] At the 2011 International Botanical Congress held in Melbourne, Australia, the decision to use the name *Acacia*, rather than the proposed *Racosperma* for this genus, was upheld.[19][20] Other *Acacia s.l.* taxa continue to be called *Acacia* by those who choose to consider the entire group as one genus.[20]

The Australian species of the genus *Paraserianthes s.l.* (namely *P. lophantha)* are deemed its closest relatives.[21] The nearest relatives of *Acacia* and *Paraserianthes s.l.* in turn include the Australian and South East Asian genera *Archidendron*, *Archidendropsis*, *Pararchidendron* and *Wallaceodendron*, all of the tribe Ingeae.[22]

List of *Acacia* species

Synonyms

List

- *Adianthum* Burm.f.
- *Chithonanthus* Lehm.
- *Cuparilla* Raf.
- *Drepaphyla* Raf.
- *Hecatandra* Raf.
- *Phyllodoce* Link
- *Racosperma* Mart.
- *Tetracheilos* Lehm.
- *Zigmaloba* Raf.

Bipinnate leaves of *Acacia dealbata*

Phyllodes of *Acacia penninervis*

Etymology

The genus name is Neo-Latin, borrowed from the Greek *akakia* , a name used by Dioscorides for some prickly species growing in Egypt, from *acacis* meaning "a point".[23] In his *Pinax* (1623), Bauhin mentioned the Greek ἀκακία from Dioscorides as the origin of the Latin name.[24]

Flowers of *Acacia retinodes*

The origin of "wattle" may be an Old Teutonic word meaning "to weave".[25] From around 700 CE, *watul* was used in Old English to refer to the flexible woody vines, branches, and sticks which were interwoven to form walls, roofs, and fences. Since about 1810 it has been used as the common name for the Australian legume trees and shrubs that can provide these branches.[25]

Species

There are about 1 080 species of *Acacia*, mostly native to Australia.[7]

Fossil record

An *Acacia*-like 14 cm (5.5 in) long fossil seed pod has been described from the Eocene of the Paris Basin.[26] *Acacia*-like fossil pods under the name *Leguminocarpon* are known from late Oligocene deposits at different sites in Hungary. Seed pod fossils of †*Acacia parschlugiana* and †*Acacia cyclosperma* are known from Tertiary deposits in Switzerland.[27] †*Acacia colchica* has been described from the Miocene of West Georgia. Pliocene fossil pollen of an *Acacia* sp. has been described from West Georgia (including Abkhazia).[28] The oldest fossil *Acacia* pollen in Australia are recorded as being from the late Oligocene epoch, 25 million years ago.[29]

Distribution and habitat

Species of *Acacia* occurs in all Australian states and territories, and on its nearby islands. About 20 species occur naturally outside Australia and 7 of these also occur in Australia. One species (*Acacia koa*) is native to Hawaii[30] and one (*Acacia heterophylla*) is native to Mauritius and Réunion in the Indian Ocean.[7][31]

They are present in all terrestrial habitats, including alpine settings, rainforests, woodlands, grasslands, coastal dunes and deserts.[9] In drier woodlands or forests they are an important component of the understory. Elsewhere they may be dominant, as in the Brigalow Belt, Myall woodlands and the eremaean Mulga woodlands.[9]

In Australia, *Acacia* forest is the second most common forest type after eucalypt forest, covering 980,000 square kilometres (378,380 sq mi) or 8% of total forest area. *Acacia* is also the nation's largest genus of flowering plants with almost 1 000 species found.[32]

Ecology

Acacia is a common food source and host plant for butterflies of the genus *Jalmenus*. The imperial hairstreak, *Jalmenus evagoras*, feeds on at least 25 acacia species.[33] Many reptiles feed on the sap, such as the native house gecko in Australia.[34] The sap is also consumed by bugs (Hemiptera), such as *Hackerobrachys viridiventris*[35] and *Sextius virescens*.[36]

Toxicity

Some species of acacia contain psychoactive alkaloids, and some contain potassium fluoroacetate.[37]

Uses

The seed pods, flowers, and young leaves are generally edible either raw or cooked.[38]

Wattle sign at Olive Pink Botanic Garden, Alice Springs, Australia (2005).

Aboriginal Australians have traditionally harvested the seeds of some species, to be ground into flour and eaten as a paste or baked into a cake. Wattleseeds contain as much as 25% more protein than common cereals, and they store well for long periods due to the hard seed coats.[39] In addition to using the edible seed and gum, the people employed the timber for implements, weapons, fuel and musical instruments.[9] A number of species, most notably *Acacia mangium* (hickory wattle), *A. mearnsii* (black wattle) and *A. saligna* (coojong), are economically important and are widely planted globally for wood products, tannin, firewood and fodder.[17] *A. melanoxylon* (blackwood) and *A. aneura* (mulga) supply some of the most attractive timbers in the genus. Black wattle bark supported the tanning industries of several countries, and may supply tannins for production of waterproof adhesives.[9]

In Vietnam, Acacia is used in plantations of non-native species that are regularly clear-cut for paper or timber uses.[40][41]

Wattle bark collected in Australia in the 19th century was exported to Europe where it was used in the tanning process. One ton of wattle or mimosa bark contained about 68 kilograms (150 pounds) of pure tannin.[42]

Cultivation

Some species of acacia – notably *Acacia baileyana*, *A. dealbata* and *A. pravissima* – are cultivated as ornamental garden plants. The 1889 publication *Useful Native Plants of Australia* describes various uses for eating.[43]

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